1. Problem Understanding

Description of the Problem

Modern healthcare is shifting from treating diseases after they occur to predicting and preventing them early. This project focuses on identifying individuals at risk of diabetes using non-invasive health data such as BMI, age, and blood pressure.

The goal is to build a reliable classification system and compare two approaches:

- Traditional Machine Learning (Logistic Regression)
- Deep Learning (Neural Networks)

This comparison helps determine which model handles data variability and noise better while predicting diabetes risk.

Importance of Predicting Diabetes Risk

Predicting diabetes risk is essential for both individuals and healthcare systems because it enables:

- **Early Intervention:** Helps detect high-risk individuals early so lifestyle changes can prevent disease progression.
- **Personalized Care & Resource Optimization:** Allows healthcare providers to focus on patients who need urgent attention.
- **Cost Reduction:** Prevents expensive complications like kidney failure or amputations.
- **Improved Quality of Life:** Supports healthier living through preventive measures.
- **Public Health Planning:** Helps design targeted awareness and screening programs.

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2. Data Exploration Insights

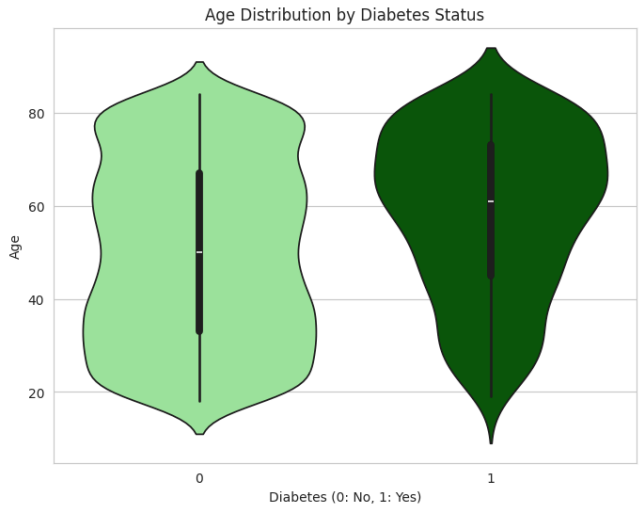
Key Observations from the Dataset

The dataset includes 6,000 patients with seven health-related features. Key findings include:

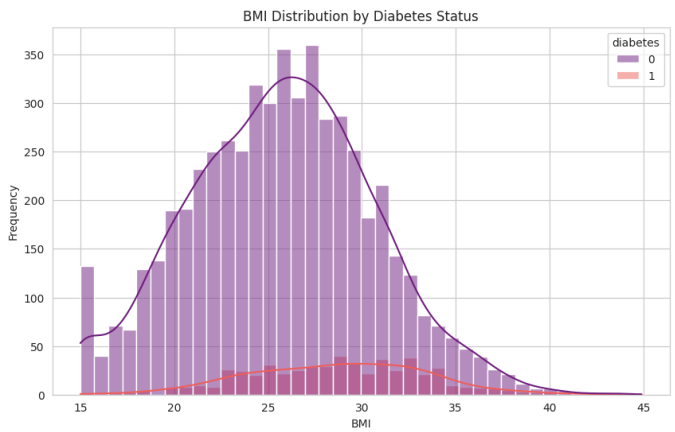
- **No Missing Data:** The dataset is clean, with complete values across all features.
- **Class Imbalance:** Only 9.03% of patients have diabetes (542 out of 6,000), which may bias the model toward predicting “no diabetes.”
- **Age Range:** Patients are between 18 and 84 years old, with an average age of 51.
- **BMI:** The average BMI is 26, indicating an overweight population.
- **Blood Pressure:** The average is 134.2 mmHg, close to hypertension levels.

Distribution Analysis

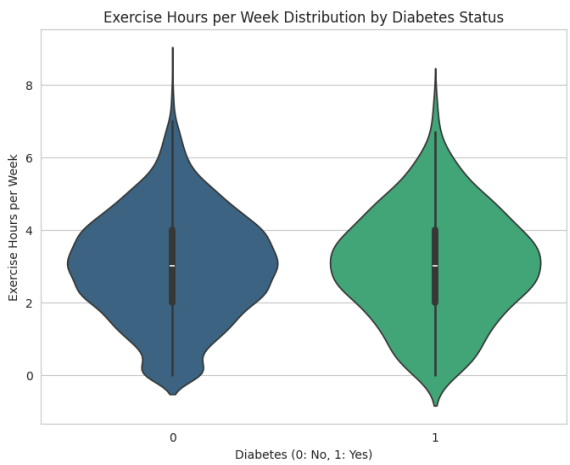
- **Age:** Diabetic patients are mostly concentrated in the 60–80 age group, making age an important factor.



- **BMI:** Higher BMI values (especially above 30) are strongly associated with diabetes risk.



- **Exercise:** Diabetic individuals generally report lower weekly exercise hours compared to non-diabetic individuals.



Important Features for Prediction

The most influential features identified are:

- **BMI (0.17 correlation):** Strongest predictor; higher BMI increases diabetes risk.
- **Blood Pressure (0.15):** Indicates metabolic health issues linked to diabetes.
- **Age (0.12):** Older individuals are more likely to be diabetic.

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3. Model Implementation

Machine Learning Model (Logistic Regression)

Logistic Regression is a linear model used for binary classification. It calculates the probability of diabetes based on weighted input features. The model provides clear coefficients that show how each feature influences the prediction.

Neural Network Model (ANN)

The deep learning model is a Sequential Artificial Neural Network with:

- **Input Layer:** 5 features (Age, BMI, Blood Pressure, Smoker, Exercise Hours)
- **Hidden Layers:** Two layers with 16 and 8 neurons using ReLU activation
- **Output Layer:** One neuron with Sigmoid activation for probability output

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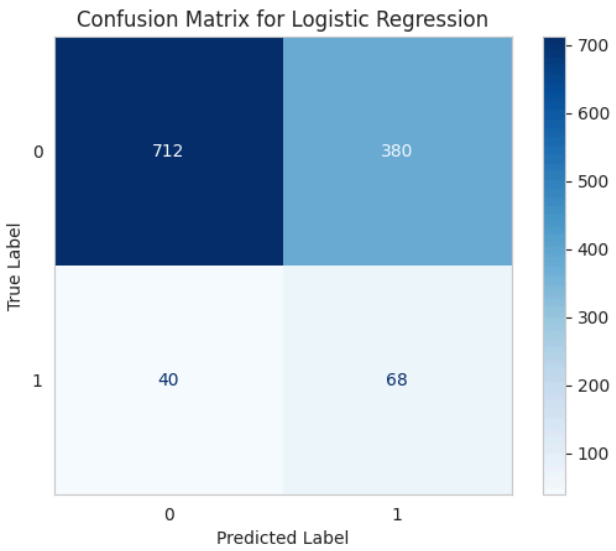
4. Model Evaluation

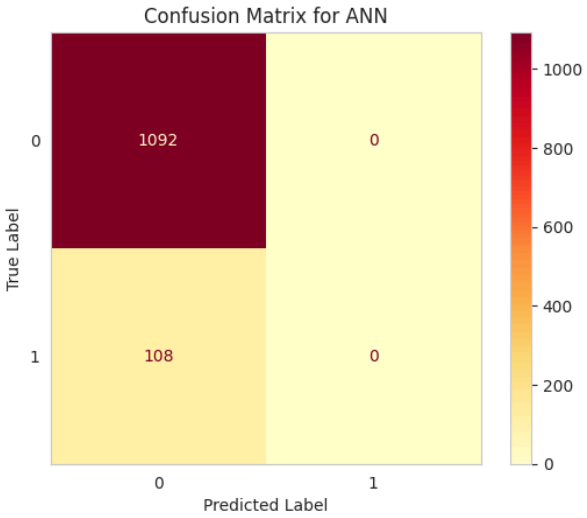
Logistic Regression Performance

- **Accuracy:** 65%
- Provides interpretable feature importance:
    - BMI (0.524) – strongest predictor
    - Age (0.337)
    - Blood Pressure (0.273)
    - Smoker (-0.176)
    - Exercise (0.033)



ANN Performance

- **Accuracy:** 90.5%
- **Loss:** 0.3473



However, performance on diabetic cases is very poor:

- **Recall (Diabetes = 1):** 0.04
- **Precision:** 0.29
- **F1-score:** 0.07

This means the model fails to correctly identify most diabetic patients.

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5. Comparative Analysis

Key Differences Between Models

- **Accuracy vs Reliability:**
  ANN shows higher accuracy (90.5%), but this is misleading due to class imbalance. Logistic Regression has lower accuracy (65%) but is more balanced.
- **Interpretability:**
  Logistic Regression clearly shows feature importance, while ANN is harder to interpret.
- **Complexity:**
  Logistic Regression is simple and linear, whereas ANN is complex and non-linear.

Reason for Accuracy Difference

The dataset is highly imbalanced (about 91% non-diabetic).

- **ANN Bias:**
  The ANN mostly predicts “no diabetes,” achieving high accuracy but missing 96% of actual diabetic cases.
- **Logistic Regression Behavior:**
  It creates a more balanced decision boundary, even though overall accuracy is lower.

Recommendations for Real-World Use

- **ANN is Not Suitable Currently:**
  With only 4% recall, it fails to detect most diabetic patients.
- **Focus on Recall Instead of Accuracy:**
  In healthcare, missing a diabetic patient (false negative) is more dangerous than a false alarm.
- **Suggested Improvements:**
  - Use class balancing techniques (SMOTE, class weights)
  - Adjust decision thresholds
  - Optimize for Recall or F1-score
- **Deployment Recommendation:**
  Logistic Regression is preferable due to its interpretability, but both models need improvement before real-world use.